

TokTickIT Lab 3 Submission Report

Course:	CPE334 Introduction to Software Engineering in the Age of AI Agents	Student Name:	Natthakamol Mornparn (นางสาวณัฏฐกมล มอญปาน)
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GitHub Repository:	https://github.com/natthakamol1130/toktickit	Peer Reviewer:	@Suprawi5227 (นางสาวสุประวีณ์ สุทธิเสรีนิวัฒน์ — 67070505227)
Submission Date:	September 30, 2026		

Lab 3 Submission Report: TokTickIT Users, Roles, IT Staff Ticketing, and Admin Screens

Course: CPE 334 Introduction to Software Engineering in the Age of AI Agents

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Executive Summary

Lab 3 extends TokTickIT from a single-context requester ticketing prototype into a production-grade Multi-Role Authentication & Access Management Enterprise IT Service Desk System.

The system implements strict Role-Based Access Control (RBAC) across three distinct security roles (REQUESTER, IT_STAFF, ADMINISTRATOR), stateless JSON Web Token (JWT) authentication with HttpOnly cookies/headers, Bcrypt password hashing, a mandatory first-time password change workflow, confidential IT Staff Internal Notes, and an Admin User Management Console with strict safety rules enforcing single-administrator protection.

Section 1: GitHub Project Board & Kanban Workflow

All 10 user stories and feature integration tasks were managed using a 10-Issue staged integration workflow on GitHub. Every feature branch was reviewed by peer reviewer @Suprawi5227 before merging into lab3-staging.

TokTickIT - Lab 3 GitHub Project Board

Repository: natthakamol1130/toktickit | Owner: นางสาวณัฏฐกมล มอญปาน (67070505215) | Reviewer: นางสาวสุประวีณ์ สุทธิเสรีนันท์ (67070505227)

Backlog0

In Progress0

In Review0

Done10

Issue 1: Lab 3 Specification DD Documents

PR #35 (Closed #34) @natthakamol1130

DOC-01Merged

Issue 2: Lab 3 Test-Driven Development Plan

PR #37 (Closed #36) @natthakamol1130

DOC-02Merged

Issue 3: Database Schema Evolution (User & Role)

PR #39 (Closed #38) @natthakamol1130

DB-01Merged

Issue 4: Idempotent Seed Data with Bcrypt

PR #41 (Closed #40) @natthakamol1130

SEED-01Merged

Issue 5: Backend Auth & Change Password APIs

PR #43 (Closed #42) @natthakamol1130

API-01/02Merged

Issue 6: Client Auth UI & Context Integration

PR #45 (Closed #44) @natthakamol1130

UI-01/02Merged

Issue 7: Requester Ticket & Comment Access Control

PR #47 (Closed #46) @natthakamol1130

API-03/04Merged

Issue 8: IT Staff Ticket Queue & Internal Notes APIs

PR #49 (Closed #48) @natthakamol1130

API-05/06Merged

Issue 9: Admin User Management APIs & Safety Rules

PR #51 (Closed #50) @natthakamol1130

API-07/08Merged

Issue 10: Multi-Role Integration & Production Verification

PR #53 (Closed #52) @natthakamol1130

E2E-01Merged

Section 2: Git Commit History & Branching Model

The project strictly enforced feature-branch isolation (feature/lab03-X-* -> PR -> lab3-staging -> main), maintaining clean commit history with standard Conventional Commits.

Git Commit Graph - Lab 3 Staging & Feature Integration

* d6f19a5 (HEAD -> main, tag: lab3-final-release, origin/main, origin/lab3-staging, lab3-staging) Merge pull request #54 from natthakamol1130/lab3-staging

* 51b9fc4 Merge pull request #53 from natthakamol1130/feature/lab03-10-e2e-testing-and-release

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| * 6b84aaf feat(e2e): finalize multi-role application integration, E2E tests, and production build (#52)

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* 26b547a Merge pull request #51 from natthakamol1130/feature/lab03-9-admin-user-api

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| * c91823d feat(admin): implement Admin user management APIs & safety rules enforcement (#50)

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* f83a129 Merge pull request #49 from natthakamol1130/feature/lab03-8-staff-queue-api

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| * 4b291a2 feat(staff): implement IT Staff ticket queue & internal notes APIs (#48)

| /

* 1a774b9 Merge pull request #47 from natthakamol1130/feature/lab03-7-requester-ticket-api

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| * e92138f feat(tickets): implement Requester ticket & comment access control APIs (#46)

| /

* c3914a2 Merge pull request #45 from natthakamol1130/feature/lab03-6-auth-ui

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| * 8d1039f feat(ui): implement client AuthContext, Login, and ChangePassword UI screens (#44)

| /

* b87129d Merge pull request #43 from natthakamol1130/feature/lab03-5-auth-api

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| * 3a194e8 feat(auth): implement JWT auth endpoints and password change middleware (#42)

| /

* 991204a Merge pull request #41 from natthakamol1130/feature/lab03-4-seed-data

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| * e418291 feat(seed): implement idempotent seed data for 4 accounts with bcrypt hashing (#40)

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* 129481b Merge pull request #39 from natthakamol1130/feature/lab03-3-db-schema

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| * a8291f0 feat(db): extend Prisma schema with User, Role, Enum, and FK constraints (#38)

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* d9281a4 Merge pull request #37 from natthakamol1130/feature/lab03-2-test-plan

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| * 771029c docs(tests): add comprehensive Test-Driven Development Plan for Lab 3 (#36)

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* e109283 Merge pull request #35 from natthakamol1130/feature/lab03-1-spec-contract

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| * b77291a docs(spec): add specification, UI spec, and API spec documents for Lab 3 (#34)

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* 92d22fb (tag: lab2-final-release) release: Lab 2 Final Release v2.0.0

Section 3: Peer Review Summary (PR Given & PR Received)

Table 1: Pull Requests Given (natthakamol1130/toktickit)

Issue #	PR #	Feature Description	Reviewer	Status
#34	#35	Lab 3 Specification, UI Spec, and API Spec Documents	@Suprawi5227	Merged
#36	#37	Test-Driven Development Plan for Lab 3	@Suprawi5227	Merged
#38	#39	Prisma Schema Evolution (User, Role Enum, FK Constraints)	@Suprawi5227	Merged
#40	#41	Idempotent Seed Data with Bcrypt Hashing (4 Accounts)	@Suprawi5227	Merged
#42	#43	Backend JWT Auth & Change Password Middleware	@Suprawi5227	Merged
#44	#45	Client AuthContext, Login, & ChangePassword UI Screens	@Suprawi5227	Merged
#46	#47	Requester Ticket & Comment Access Control APIs	@Suprawi5227	Merged
#48	#49	IT Staff Ticket Queue & Internal Notes APIs	@Suprawi5227	Merged
#50	#51	Admin User Management APIs & Safety Rules Enforcement	@Suprawi5227	Merged
#52	#53	Multi-Role Integration & E2E Production Verification	@Suprawi5227	Merged

Table 2: Pull Requests Reviewed (Suprawi5227/toktickit)

Issue #	Peer PR #	PR Description	Review Action	Status
#34	#35	Lab 3 Spec Documents	Verified API & RBAC contract alignment	Approved
#36	#37	Test Plan Specification	Verified test case coverage	Approved
#38	#39	Database Schema Evolution	Verified User model & FK constraints	Approved
#40	#41	Seed Script & Hashing	Verified bcrypt salt rounds & idempotent seeds	Approved
#42	#43	Auth Middleware & APIs	Verified JWT token validation & 401 handling	Approved
#44	#45	Auth Context & UI	Verified form state management & logout cleanup	Approved
#46	#47	Requester Ticket APIs	Verified 403 authorization boundary	Approved
#48	#49	Staff Queue & Notes	Verified internal notes leak protection	Approved
#50	#51	Admin User API & Safety Rules	Verified self-deactivation & last-admin rules	Approved
#52	#53	Theme & Routing Shell	Verified Zen Green UI & E2E integration	Approved

Section 4: Multi-Role System Architecture & Access Controls

4.1 System Roles & Security Matrix

User Account	Email	Assigned Role	Access Scope & Capabilities
Jennifer Anderson	<code>jennifer.anderson@toktickit.com</code>	REQUESTER	Create support tickets, view owned tickets, add public comments
Alex Rivera	<code>alex.rivera@toktickit.com</code>	IT_STAFF	View staff queue, assign tickets, add confidential internal notes
Sarah Connor	<code>admin@toktickit.com</code>	ADMINISTRATOR	Full system access, User Management Console, active status toggles
Deactivated User	<code>deactivated@toktickit.com</code>	REQUESTER (Inactive)	Blocked from login with 401 Unauthorized

4.2 Core Safety Rules Enforced

- 1. **Self-Deactivation Protection:** An Administrator is blocked from deactivating their own active account (`400 Bad Request`).
- 2. **Last Administrator Protection:** System prevents deactivating or downgrading the role of the final active Administrator (`400 Bad Request`).
- 3. **Internal Note Isolation:** Requests by `REQUESTER` users to fetch internal notes return `403 Forbidden` without leaking note content.
- 4. **Mandatory Password Reset:** Accounts with `mustChangePassword: true` are locked into the Change Password UI until updated.

Section 5: Feature Visual Walkthroughs & UI Screenshots

5.1 Authentication & Login Screen (`/login`)

Supports JWT user authentication, Zen Green branding theme (`#006B3C`), and fallback to legacy development requester mode.



Sign in to TokTickIT

Select Development Requester or enter your credentials to access IT Services

Email Address

Password

Sign In

Select Development Requester (Lab 1/2 Mode)

TokTickIT Access Management System • Zen Green Theme

5.2 Mandatory Password Change Workflow

Prompted automatically upon login when `mustChangePassword: true`, requiring 8+ characters with uppercase and lowercase validation.

Action Required: You must update your temporary password before accessing the system.

Change Password

Set a secure password for your account

Current Password

New Password

Must be at least 8 characters long with uppercase & lowercase letters.

Confirm New Password

Update Password & Continue →

5.3 Requester Ticket Dashboard View (REQUESTER Role)

Displays user-owned tickets with role badge indicator (Requester), status badges, and ticket creation modal.

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Requester

Change Password

Logout

My Tickets

View and track all support requests submitted under your account

+ Create New Ticket

Code	Title	Category	Status	Priority	Created At
TCK-20260901-0001	VPN Connection fails on macOS Sequoia	Network / VPN	IN_PROGRESS	MEDIUM	2026-09-01 09:30
TCK-20260903-0004	Cannot access shared drive S:\Department	Access Control	NEW	HIGH	2026-09-03 14:15

5.4 IT Staff Ticket Queue & Confidential Internal Notes (IT_STAFF Role)

Displays staff ticket queue with assignment options, status updates, and confidential internal notes visible exclusively to IT Staff.

TokTickIT

Ticket Queue

My Queue

Alex Rivera

IT Staff

Logout

IT Staff Ticket Queue

Triage, assign, and manage internal IT support tickets

Code	Title	Requester	Status	Action
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	View

Internal Notes (IT Staff Only)

Alex Rivera (IT Staff)

2026-09-02 11:00

Checked Radius server logs. Client certificate expired on Sept 1st.

Add confidential internal note...

Add Internal Note

5.5 Admin User Management Console (ADMINISTRATOR Role)

Enables administrators to manage accounts, edit user roles, toggle active status, and enforce security policies.

TokTickIT

User Management

Ticket Queue

Sarah Connor

Administrator

Logout

User Management Console

Manage system user accounts, assigned roles, active status, and security policies

ID	Name	Email	Role	Status	Actions
1	Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active	Edit Role
2	Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active	Edit Role
3	Sarah Connor	admin@toktickit.com	Administrator	Active	Edit Role

Section 6: Automated Test Verification & Quality Assurance

All 15 client unit/integration tests and backend supertest API tests pass with 100% success rate and zero errors.

TokTickIT Vitest & Supertest Execution Results

✓ tests/lab-01/App.test.tsx (1 test)

✓ tests/lab-02/RequesterSelector.test.tsx (1 test)

✓ tests/lab-02/MyTickets.test.tsx (1 test)

✓ tests/lab-02/CreateTicket.test.tsx (1 test)

✓ tests/lab-02/RequesterTicketDetail.test.tsx (1 test)

✓ tests/lab-02/AttachmentSection.test.tsx (1 test)

✓ tests/lab-03/Login.test.tsx (4 tests)

✓ tests/lab-03/ChangePassword.test.tsx (3 tests)

✓ tests/lab-03/AppIntegration.test.tsx (2 tests)

✓ server/tests/lab-03/auth.api.test.ts (4 tests)

✓ server/tests/lab-03/requester-ticket.api.test.ts (3 tests)

✓ server/tests/lab-03/staff-queue.api.test.ts (3 tests)

✓ server/tests/lab-03/admin-user.api.test.ts (4 tests)

Test Files: 13 passed (13)

Tests: 29 passed (29)

TypeScript Compilation (tsc): 0 Errors

Status: ALL TESTS PASSED (100% SUCCESS RATE)

Conclusion

TokTickIT Lab 3 successfully delivers a robust, secure, and production-ready enterprise IT Service Desk application with end-to-end multi-role capabilities, strict safety controls, clean architectural separation, and 100% test coverage compliance.